

Practicals\Task5.cpp

```
1  #include <iostream>
2  #include <vector>
3  #include <cmath>
4  #include <cstdlib>
5  using namespace std;
6
7  class min_distance
8  {
9      private:
10         int x1;
11         int x2;
12         int y1;
13         int y2;
14
15     public:
16         min_distance() = default;
17
18         min_distance(int x1,int x2,int y1,int y2)
19         {
20             this->x2 = x2 ;
21             this->y1 = y1 ;
22             this->y2 = y2 ;
23             this->x1 = x1 ;
24
25             float X = (x2 - x1) * (x2 - x1) ;
26             float Y = (y2 - y1) * (y2 - y1) ;
27             float D = X + Y ;
28             cout << "Shortest distance between point (x1,y1) and (x2,y2) ie (" << x1 << ","
<< y1 << ") and (" << x2 << "," << y2 << ") is : " << sqrt(D) << endl ;
29         }
30
31         int get_x1()
32         {
33             cout << "Enter value of point x1 : " ; cin >> x1 ;
34             return x1 ;
35         }
36
37         int get_y1()
38         {
39             cout << "Enter value of point y1 : " ; cin >> y1 ;
40             return y1 ;
41         }
42
43         int get_x2()
44         {
45             cout << "Enter value of point x2 : " ; cin >> x2 ;
46             return x2 ;
47         }
```

```
48
49     int get_y2()
50     {
51         cout << "Enter value of point y2 : " ; cin >> y2 ;
52         return y2 ;
53     }
54
55     float shord(bool msg=true)
56     {
57         float x1 = get_x1() ;
58         float y1 = get_y1() ;
59         float x2 = get_x2() ;
60         float y2 = get_y2() ;
61
62         float X = (x2 - x1) * (x2 - x1) ;
63         float Y = (y2 - y1) * (y2 - y1) ;
64         float DI = X + Y ;
65         if(msg == true)
66         {
67             cout << "Shortest distance between point (x1,y1) and (x2,y2) ie (" << x1 <<
68             ", " << y1 << ") and (" << x2 << ", " << y2 << ") is : " << sqrt(DI) << endl ;
69         }
70         return sqrt(DI);
71     }
72 };
73
74 int main ()
75 {
76     system("cls");
77     cout << string(84,'=') << endl ;
78     min_distance s1(3,5,4,7);
79
80     min_distance s2;
81     cout << string(84,'=') << endl ;
82     float distance = s2.shord();
83     cout << string(84,'=') << endl ;
84
85 }
```